Symmetries in Physics — Exercise Sheet 2

Structure of groups, quotients, and products

Variant: Questions only

Exercise 1. Let $H \leq G$ be a subgroup of index 2.

(i) Show that $g^2 \in H$ for every $g \in G$.

(ii) Deduce that A_4, of order 12, has no subgroup of order 6. This is the counterexample to the converse of Lagrange's theorem that Lecture 1 states without proof.

Exercise 2. Compute the conjugacy classes of (i) D_6 and (ii) S_4, and verify the class equation $|G| = \sum_i |C_i|$ in both cases.

Exercise 3. Use the first isomorphism theorem to identify the following quotients:

(i) $\mathrm{GL}(n,\mathbb{R})/\mathrm{SL}(n,\mathbb{R})$;

(ii) $(\mathbb{C}\setminus\{0\})/\mathrm{U}(1)$;

(iii) $\mathrm{U}(1)/\{\pm 1\}$.

Exercise 4. Direct products are defined in the reading.

(i) Show that $C_4 \times C_6 \cong C_2 \times C_{12}$.

(ii) Show that $C_4 \times C_6$ is not isomorphic to $C_2 \times C_2 \times C_6$.

Exercise 5.

(i) Show that $D_4/Z(D_4)$ is isomorphic to the Klein four-group.

(ii) Using Exercise 2, find $Z(D_6)$ and show that $D_6/Z(D_6) \cong D_3$. So the quotient by the centre need not be abelian.

Exercise 6 (semidirect products). Let $N \trianglelefteq G$ and $H \leq G$ satisfy $N \cap H = \{e\}$ and $NH = G$.

(i) Show every $g \in G$ is uniquely $g = nh$ ($n \in N$, $h \in H$), and that the product reads

$$(n_1 h_1)(n_2 h_2) = \left(n_1\, h_1 n_2 h_1^{-1}\right)(h_1 h_2)$$

— the *(internal) semidirect product* $N \rtimes H$, with H acting on N by conjugation.

(ii) Show that $S_n \cong A_n \rtimes C_2$ for $n \geq 2$.

(iii) Show that C_4 is *not* a semidirect product of two nontrivial subgroups.

Exercise 7 (composition series). As in Lecture 2, a *composition series* of a finite group G is a chain

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_k = \{e\},$$

in which each quotient G_i/G_{i+1} is simple.

(i) Find a composition series of S_4 and list its factors.

(ii) Find the composition series of S_3 and of C_6, and compare their factors. What does this say about how much the factors know about the group?